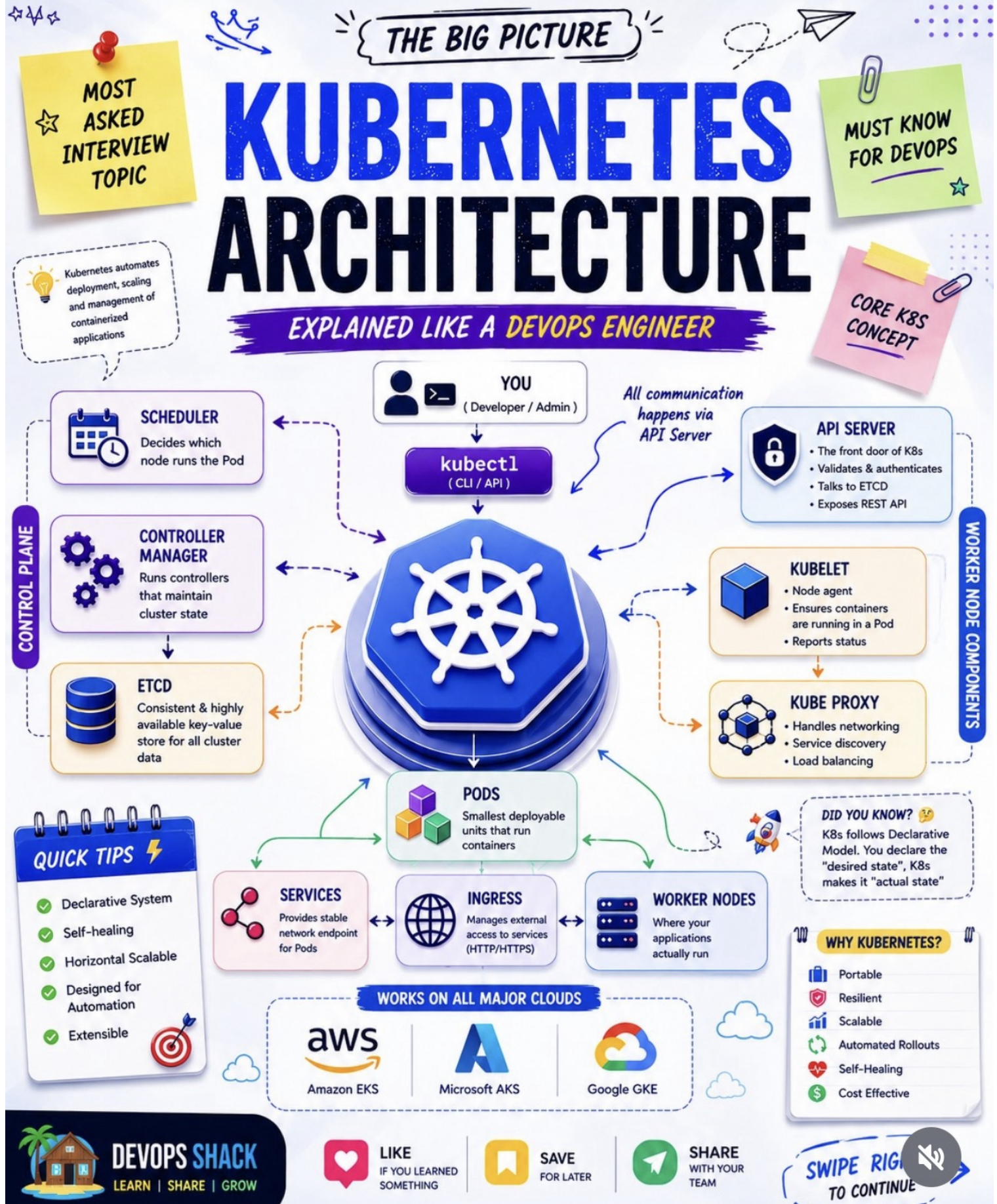
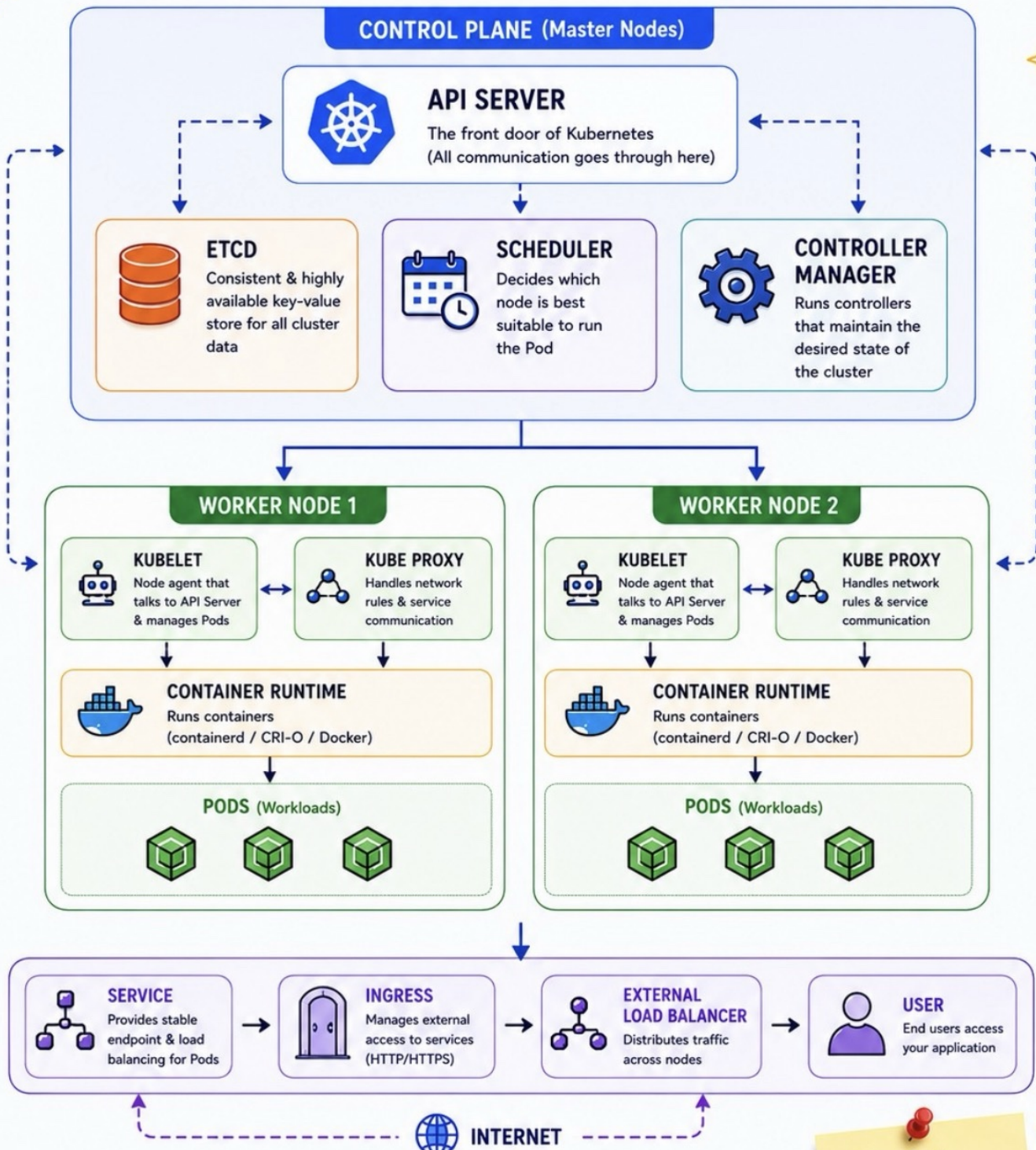




HOW KUBERNETES ACTUALLY WORKS



■ Control Plane (Brain)
 ■ Worker Nodes (Muscles)
 ■ Networking (Connects)
 ■ Storage (Data)

Remember:
Control Plane makes decisions.

Worker Nodes run your applications.

CONTROL PLANE

The Brain of Kubernetes



CONTROL PLANE COMPONENTS



API SERVER

The front door of Kubernetes.
All communication goes through the API Server.



ETCD

Consistent and highly available key-value store for all cluster data.

Stores the state of the entire cluster.



SCHEDULER

Decides which node is best suitable to run a new Pod.

Looks at resources, constraints & policies.



CONTROLLER MANAGER

Runs controllers that maintain the desired state of the cluster.

Example: Node, ReplicaSet, Deployment, etc.

WHAT IT DOES



Validates & processes all API requests



Stores & retrieves cluster state



Schedules workloads efficiently



Ensures the actual state matches the desired state



Continuously monitors & reconciles

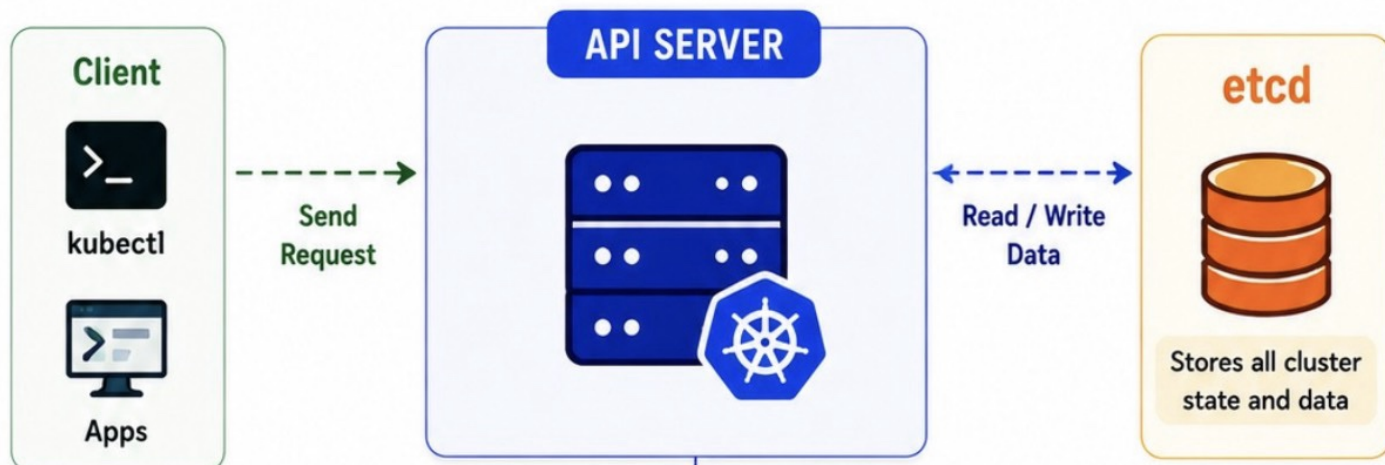


KEY TAKEAWAY: The Control Plane makes global decisions for the cluster. It does not run your applications.



API SERVER

The Front Door of Kubernetes



WHAT IT DOES



AUTHENTICATION

Verifies who you are
(Bearer Token,
Certificates, etc.)



AUTHORIZATION

Checks what you are
allowed to do
(RBAC, Policies)



VALIDATION

Validates the request
data before processing
it further



REST API

Exposes Kubernetes
resources via
RESTful API



CLUSTER COMMUNICATION

Acts as the central
hub for all cluster
communication

REQUEST FLOW



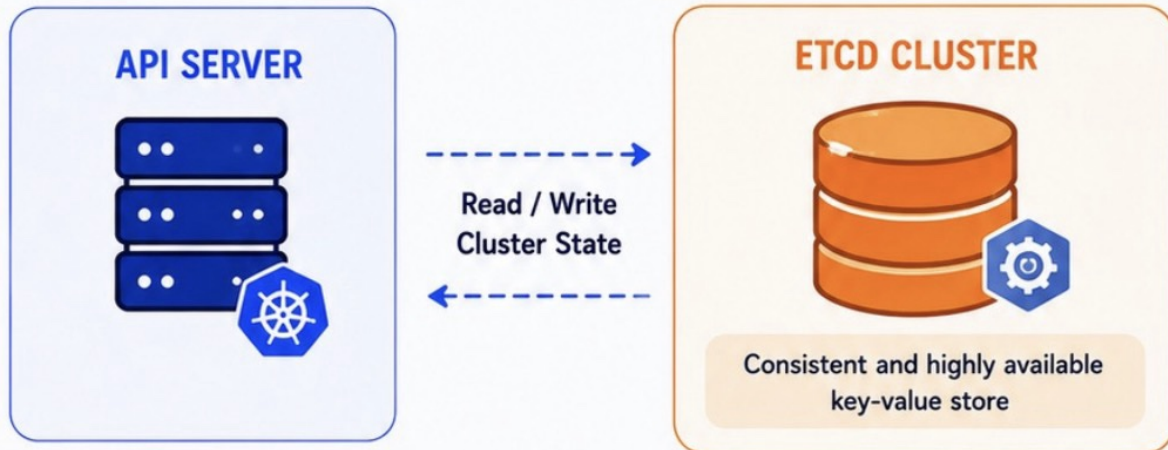
KEY TAKEAWAY

The API Server is the **gateway** to the Kubernetes cluster.
Everything in Kubernetes goes through the API Server.



ETCD

The *Heart* of Kubernetes



WHAT IT STORES



PODS

Information about all Pods



NODES

Information about all Nodes



SECRETS

All Secrets and credentials



DEPLOYMENTS

Desired state of Deployments



SERVICES

ClusterIP, Endpoints, Service definitions



CONFIGMAPS

Configuration data

In short: Everything that defines the desired and current state of your cluster.

KEY CHARACTERISTICS



CONSISTENT

Strong consistency ensures cluster state is reliable



HIGHLY AVAILABLE

Runs in a cluster (odd number of nodes) to avoid downtime



DURABLE

Data is persisted on disk and survives failures



KEY-VALUE STORE

Simple, fast and optimized for storing cluster state



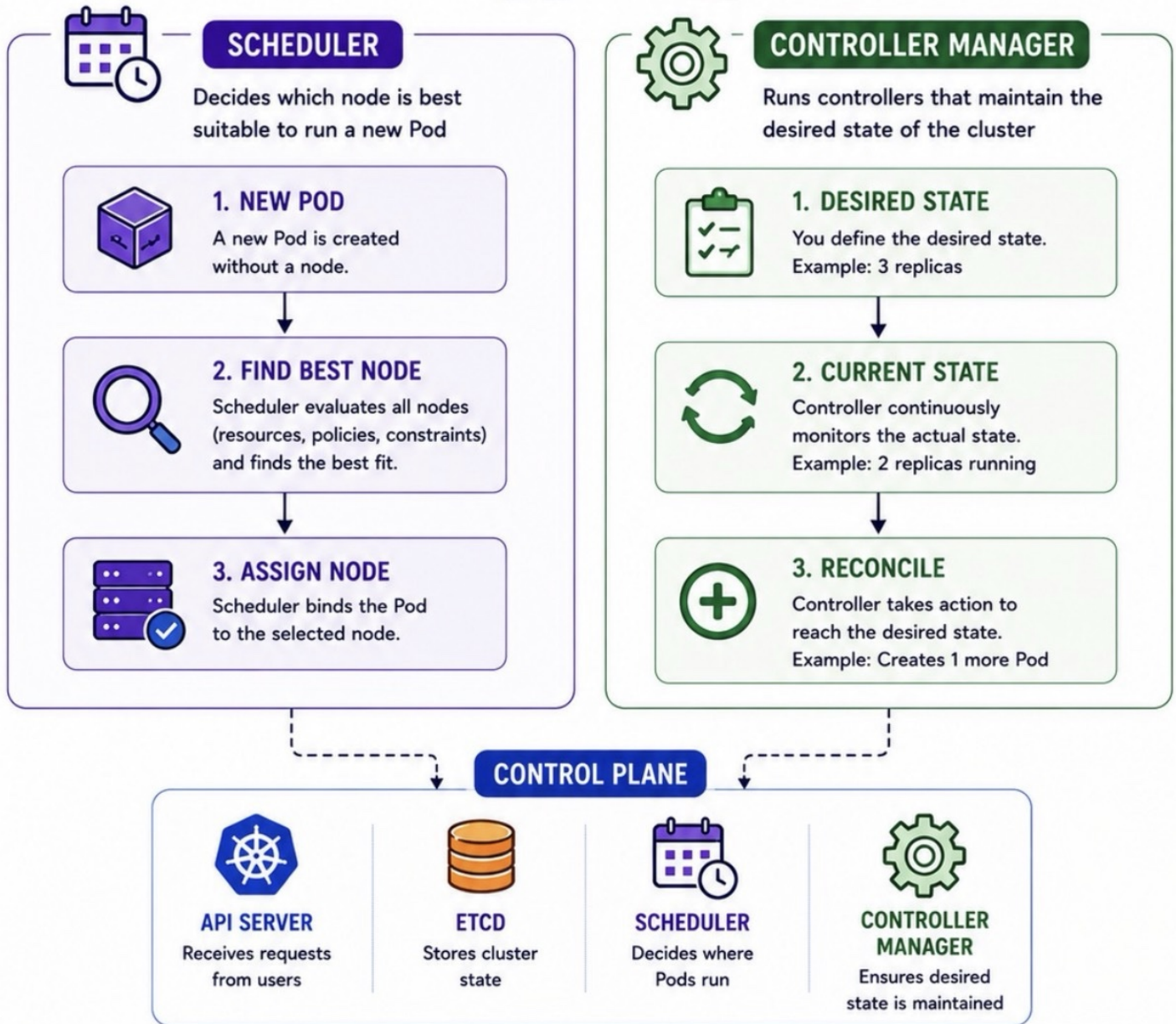
KEY TAKEAWAY

etcd is a distributed key-value database that stores all cluster data. Without etcd, Kubernetes has **no memory**.



SCHEDULER & CONTROLLER MANAGER

The Decision Makers



KEY TAKEAWAY



Scheduler makes placement decisions.



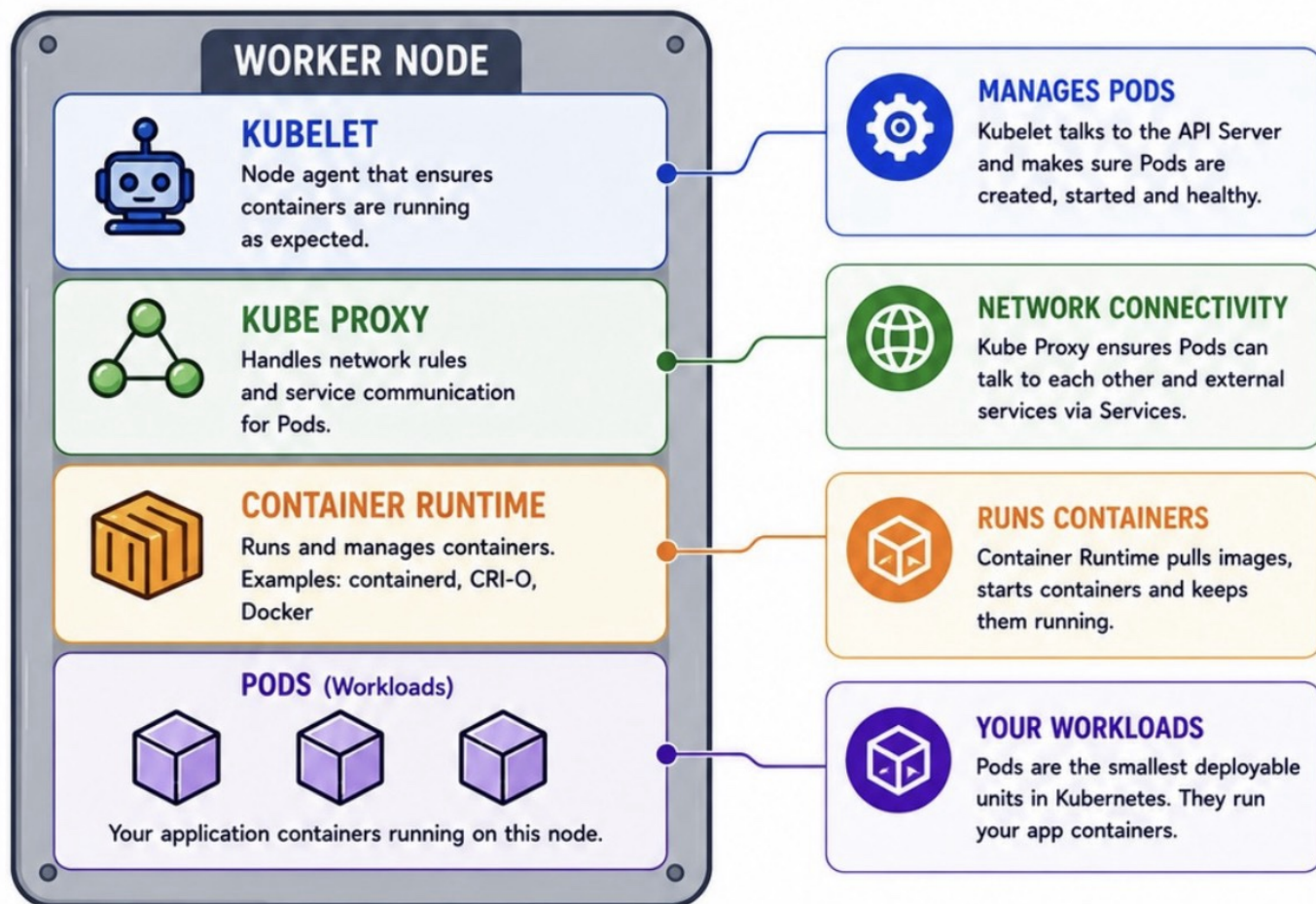
Controller Manager makes sure the cluster stays in the desired state.



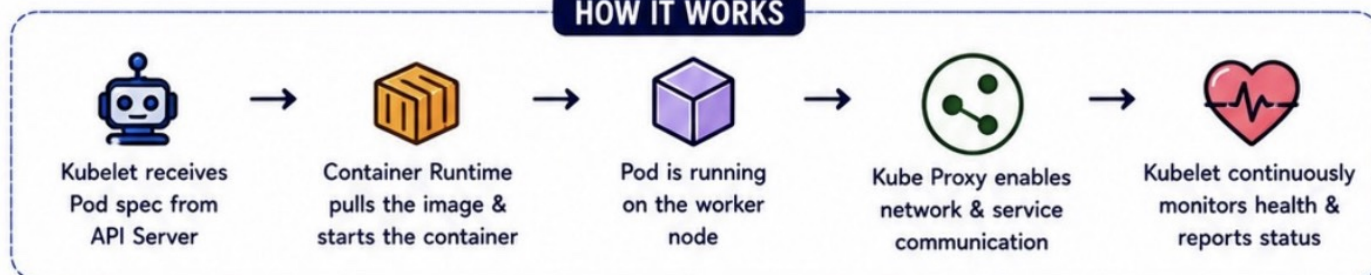
WORKER NODE ARCHITECTURE

The Machines That *Run* Your Workloads

7/10



HOW IT WORKS



KEY TAKEAWAY

Worker Nodes do the *heavy lifting*.
They run your containers and keep everything healthy.



KUBERNETES NETWORKING



How Traffic Flows in a Cluster

